

# **Autonomy of Unmanned Aerial Vehicles**

## **Introduction to Inverted Pendulum**

### **Part A. State Space Modeling**

*by*

Dr. Carlos Sotelo – Dr. David Sotelo

# Context

It is an **advanced level course in control engineering**, in which the student **implements** their own **control algorithms**, applying the competencies developed in the focus stage of their study plan.

- It **requires prior knowledge** of modeling and automation and control system design.
- The student **supports his proposal** by the analysis of the performance of the control system.

QDrone 2

Qube-Servo 3

Aero 2



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Configuration 1



Qube-Servo 3



Configuration 2



# Context

## DC Motor

It is supplied with two standard items:

- Inverted pendulum

The inverted pendulum is mounted to the equipment with magnets. Here, the encoder provide an additional sensor input to the system [6].

- Inertia disk

A small aluminum part is mounted to the equipment using magnets.



Qube-Servo 3



Fig. 1 Inverted pendulum.



Fig. 2 Inertia disk.

**Objective:** To obtain and validate the linear time invariant state space representation for the Pendulum DOWN / Crane Configuration using Real-Time Hardware-in-the-Loop Simulation by Quanser Interactive Labs.

# Practice 09

## Part A. Introduction

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### Abstract

The rotatory inverted pendulum, known as Furuta pendulum, is a system used in control labs that provides a compact yet impressive platform for control demonstrations and draws the attention of the students to develop their own control laws. It consists of a driven arm which rotates in the horizontal plane and a pendulum attached to that arm which is free to rotate in the vertical plane. In this problem situation, you must obtain and validate the linear time invariant state space representation for the Pendulum DOWN / Crane Configuration.

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### Topics Covered

- First principles modeling
- State space representation
- Model validation

### Prerequisites

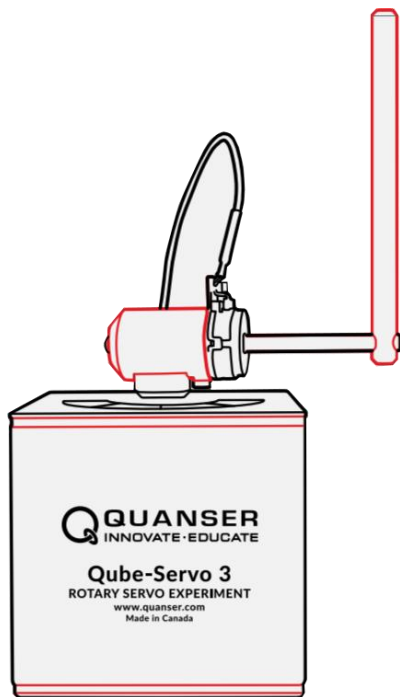
- Hardware Interfacing laboratory experiment
- Rotary Pendulum Modeling laboratory experiment



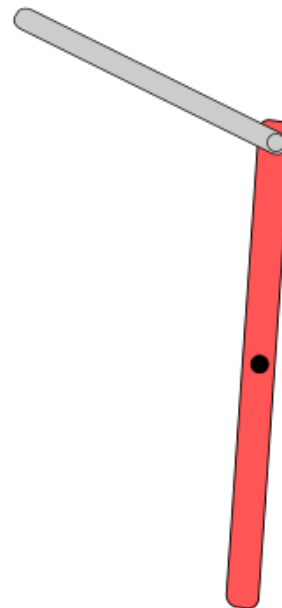
# Practice 09

## Part A. Context

The rotary pendulum is shown in Fig. 1.1. Additionally, the model is presented in Fig. 1.2:



**Fig. 1.1** Qube-Servo 3 Inverted pendulum.



**Fig. 1.2** Qube-Servo 3 Inverted pendulum  
DOWN / Crane Configuration.



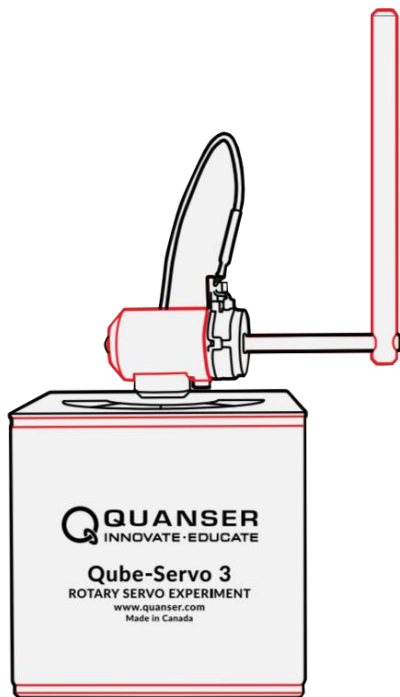
### Considering:

- The rotary arm pivot is attached to the Qube Servo 3.

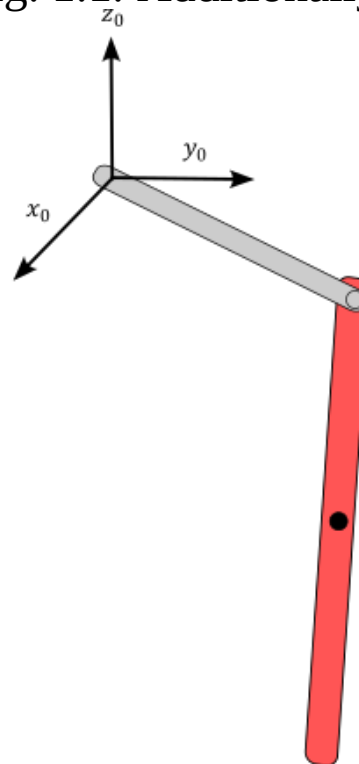
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### Considering:

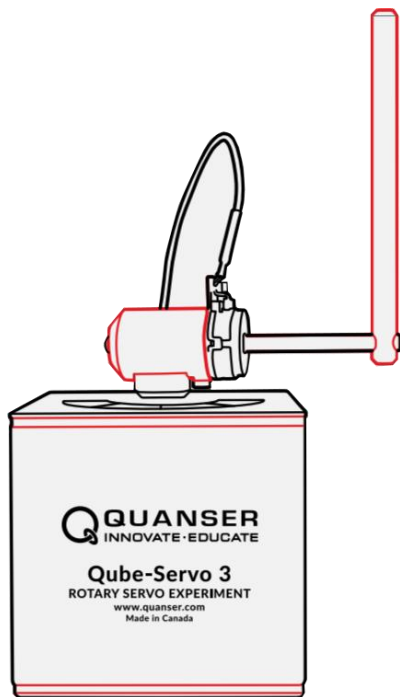
➤ The rotary arm pivot is attached to the Qube Servo 3. The arm has:



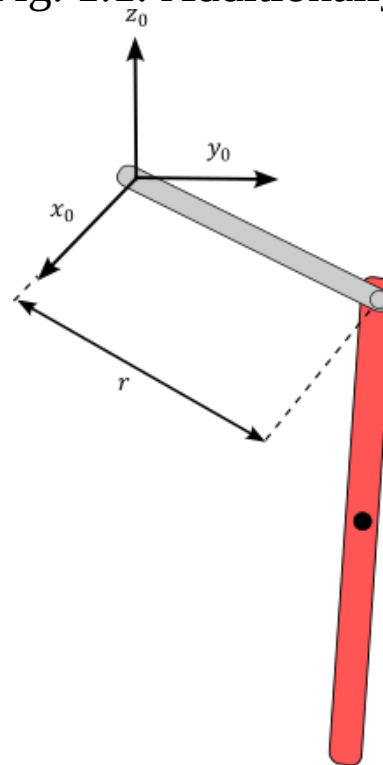
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DOWN / Crane Configuration.



### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:  
a length of  $r$ ,

# Practice 09

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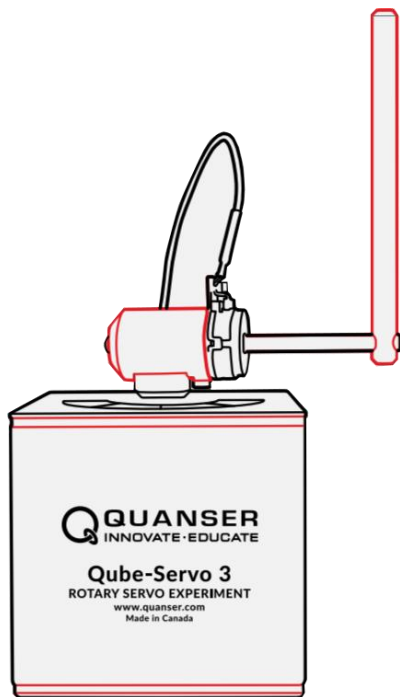


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

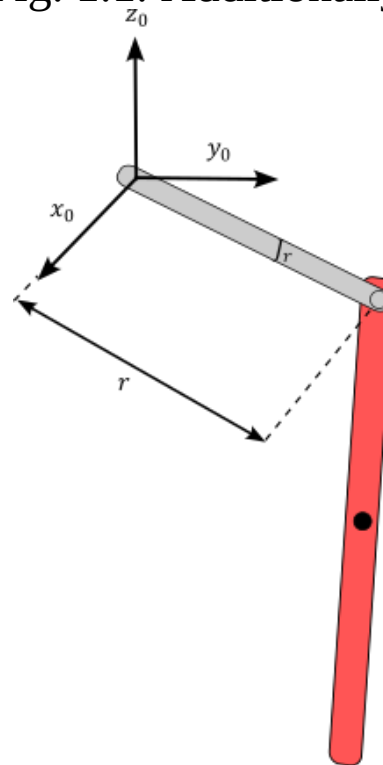


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:



- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
  - a length of  $r$ ,
  - a moment of inertia of  $J_r$ ,

# Practice 09

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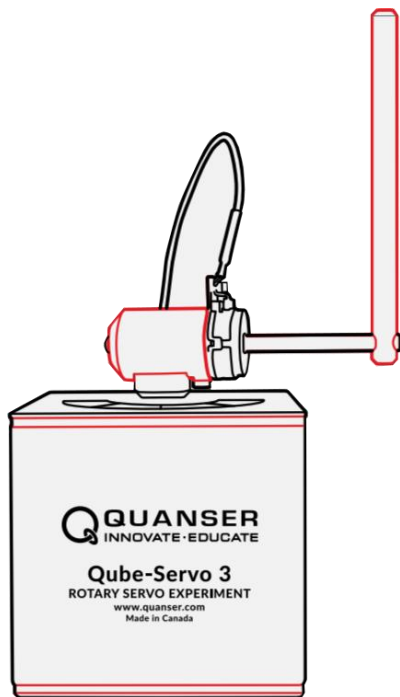


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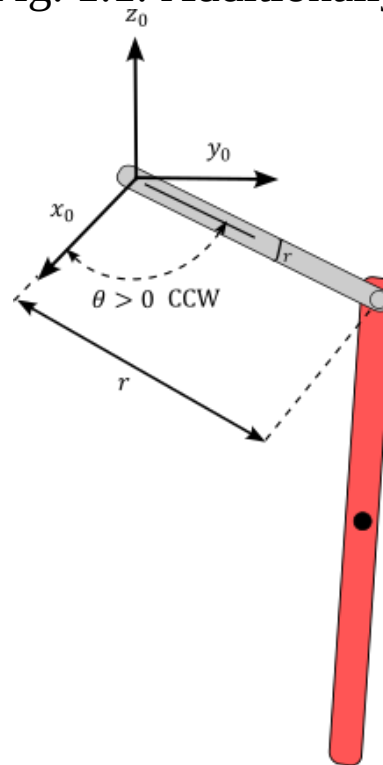


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
  - a length of  $r$ ,
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- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).



# Practice 09

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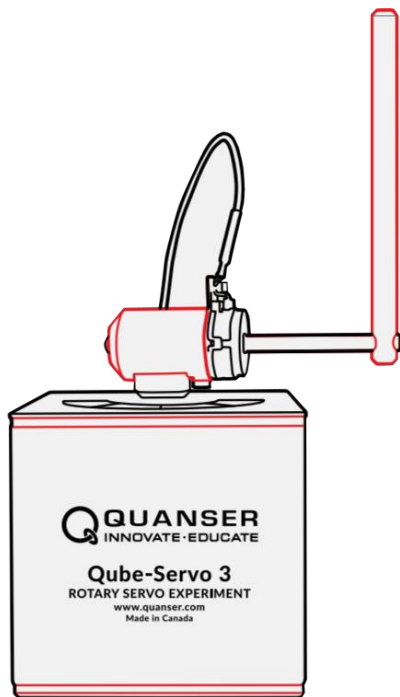


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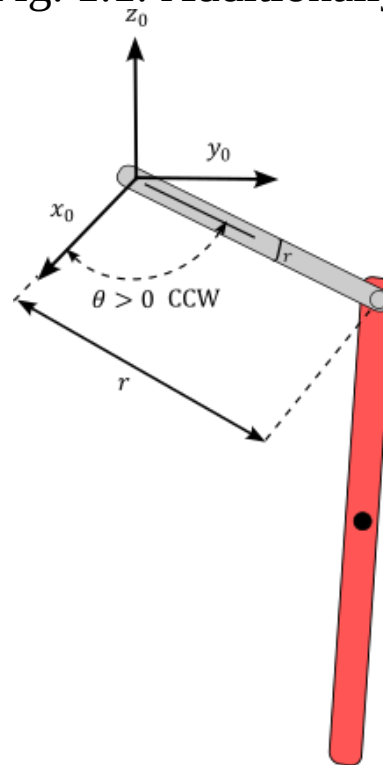


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
  - a length of  $r$ ,
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- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .



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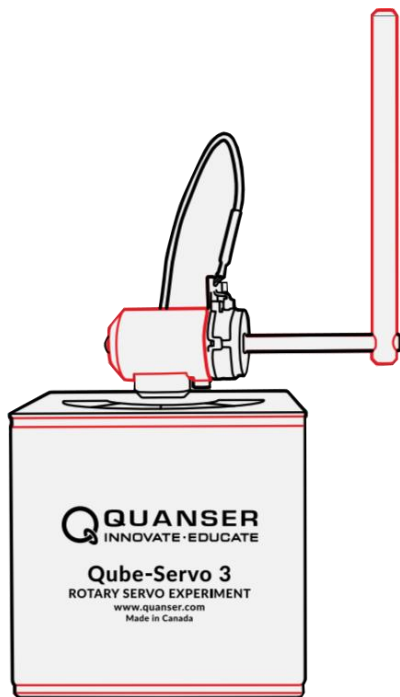


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

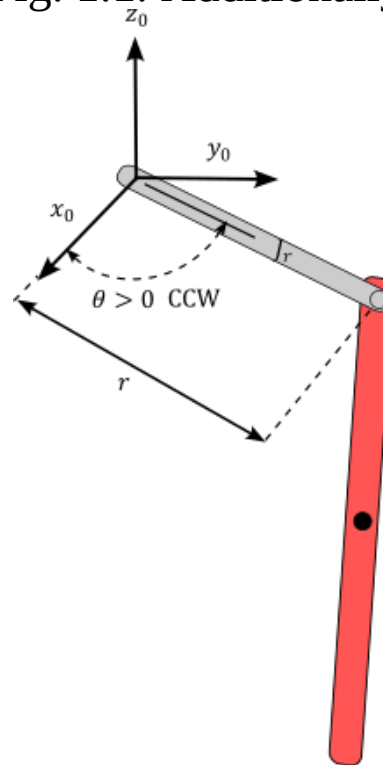


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
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  - a moment of inertia of  $J_r$ ,
- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .
- The pendulum link is connected to the end of the rotary arm. It has:



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## Part A. Context

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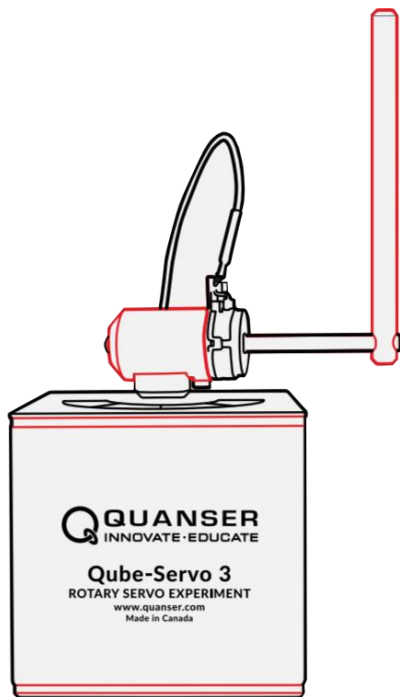


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

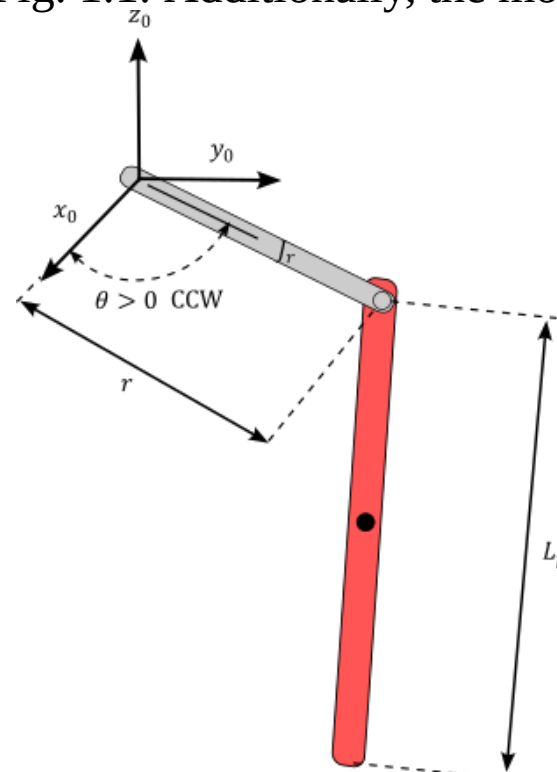


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
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- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .
- The pendulum link is connected to the end of the rotary arm. It has:
  - a total length of  $L_p$ ,



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## Part A. Context

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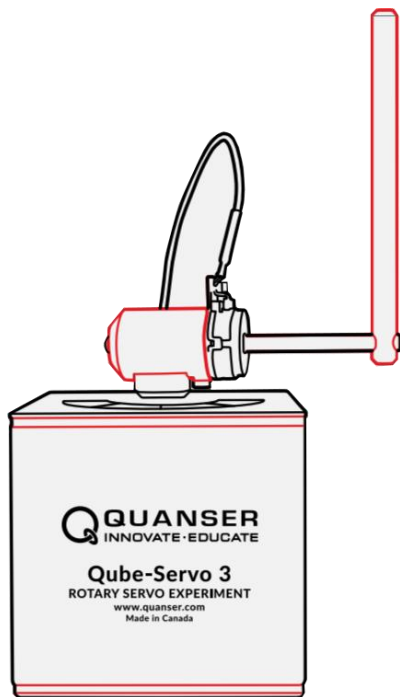


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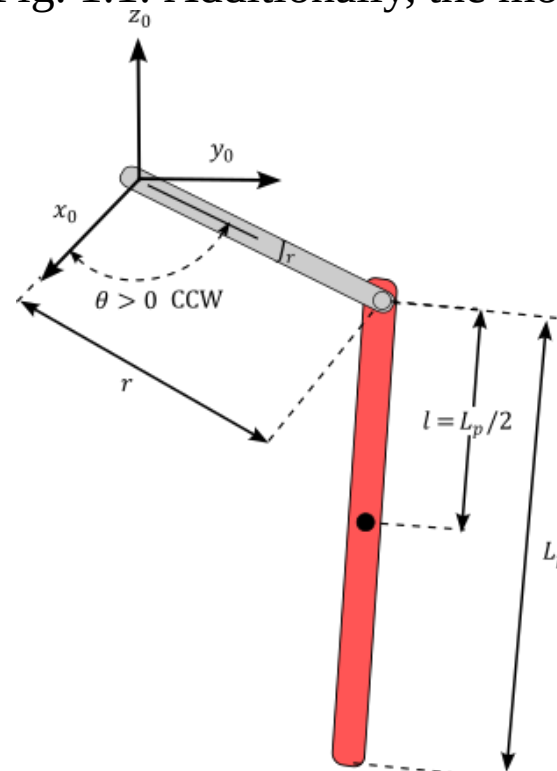


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
  - a length of  $r$ ,
  - a moment of inertia of  $J_r$ ,
- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .
- The pendulum link is connected to the end of the rotary arm. It has:
  - a total length of  $L_p$ ,
  - the center of mass is at  $l = L_p/2$ ,



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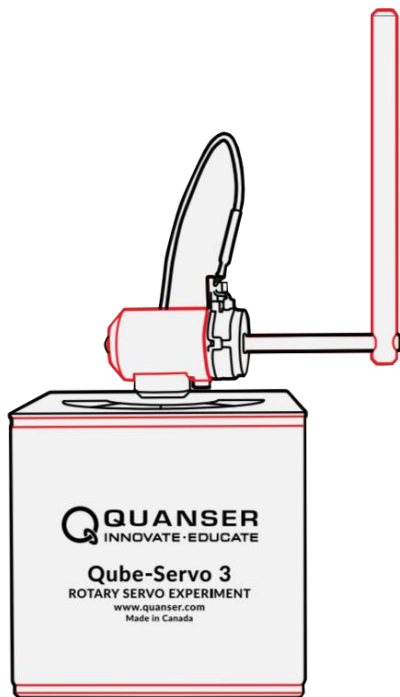


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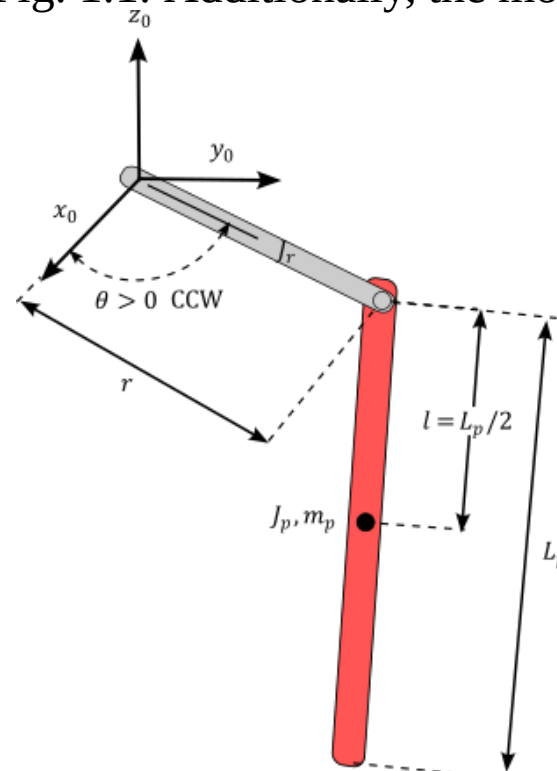


Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

### Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
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  - a moment of inertia of  $J_r$ ,
- Its angle  $\theta(t)$  increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .
- The pendulum link is connected to the end of the rotary arm. It has:
  - a total length of  $L_p$ ,
  - the center of mass is at  $l = L_p/2$ ,
  - the moment of inertia about its center of mass is  $J_p$ ,





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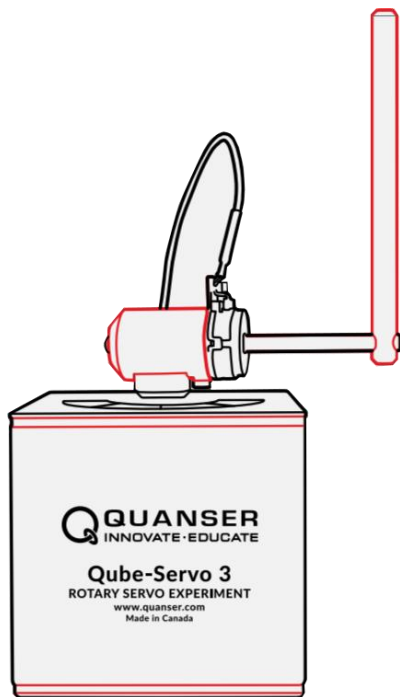


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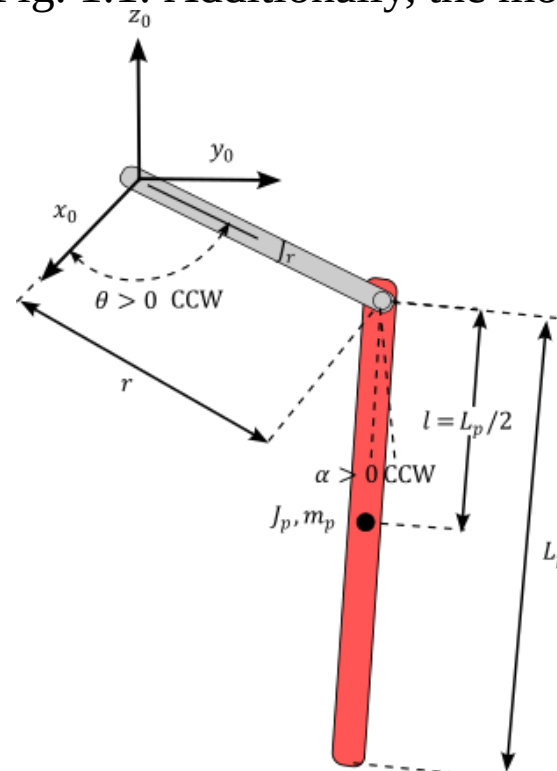


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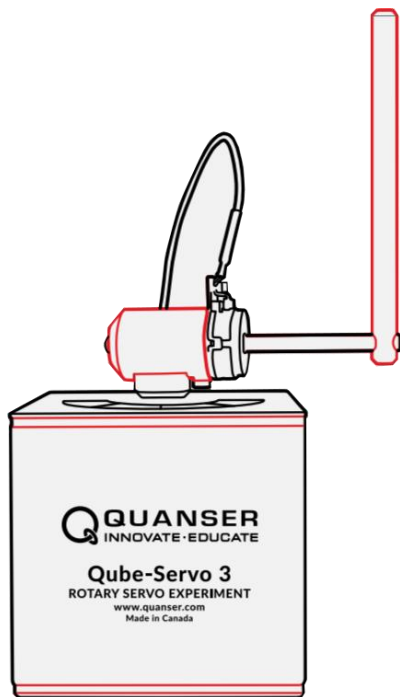
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- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive,  $v_m(t) > 0$ .
- The pendulum link is connected to the end of the rotary arm. It has:
  - a total length of  $L_p$ ,
  - the center of mass is at  $l = L_p/2$ ,
  - the moment of inertia about its center of mass is  $J_p$ ,
- The rotary pendulum angle  $\alpha(t)$  is zero when it is hanging downward and increases positively when rotated CCW.



# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum



- ① Analyze the differential equation that describes the process behavior of the system.
- ② Obtain the linear time invariant state space representation.
- ③ Validate your model using Quanser Interactive Labs.

Fig. 1.1 Qube-Servo 3 Inverted pendulum.

# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

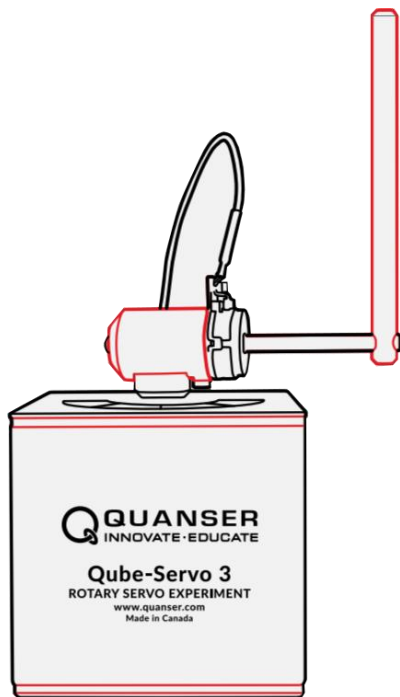


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.

The equations of motion (EOM) for the pendulum system were developed using the Euler Lagrange method. The resultant nonlinear EOM are:

$$\begin{aligned} J_r \ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) &= \tau - b_r \dot{\theta}(t) & \tau &= \frac{k_m}{R_m} [v_m(t) - k_m \dot{\theta}(t)] \\ J_p \ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) &= -b_p \dot{\alpha}(t) \end{aligned}$$

where:

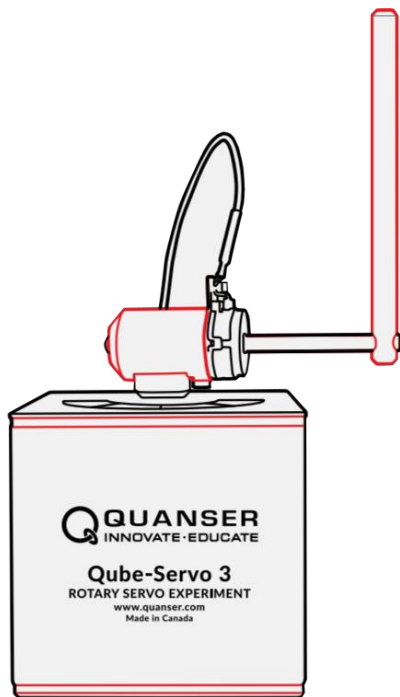
- $J_r = m_r r^2 / 3$  is the moment of inertia of the rotary arm with respect to the pivot (i.e. rotary arm axis of rotation)
- $J_p = m_p L_p^2 / 3$  corresponds to the moment of inertia of the pendulum link relative to the pendulum pivot (i.e. axis of rotation of pendulum)
- $b_r$  represents the viscous damping acting on the rotary arm
- $b_p$  stands for the viscous damping acting on the pendulum link
- $\tau$  is the applied torque at the base of the rotary arm

- 2 Obtain the linear time invariant state space representation.

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## Part A. State Space Modeling

### Example: Inverted Pendulum



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Fig. 1.1 Qube-Servo 3 Inverted pendulum.

# Practice 09

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### Example: Inverted Pendulum

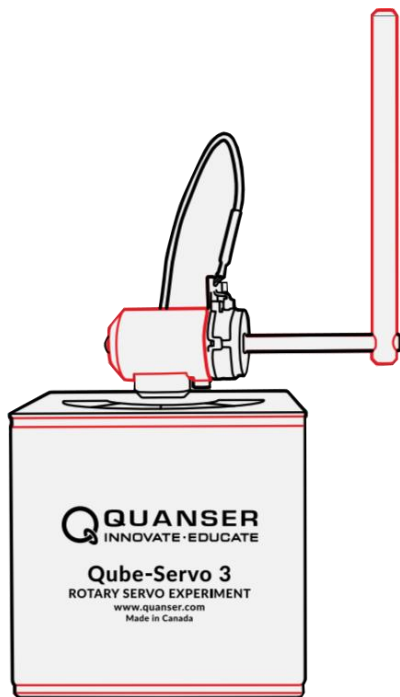


Fig. I.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.
- 2 Obtain the linear time invariant state space representation.

You must consider that based on model, we have:

#### States

$$\mathbf{x}(t) = \begin{bmatrix} \theta(t) \\ \alpha(t) \\ \dot{\theta}(t) \\ \dot{\alpha}(t) \end{bmatrix}; \begin{array}{l} \text{angle position of the arm} \\ \text{angle position of the pendulum} \\ \text{angular velocity of the arm} \\ \text{angular velocity of the pendulum} \end{array}$$

#### Inputs

$$\mathbf{u}(t) = [v_m(t)]; \text{ motor voltage}$$

#### Outputs

$$\mathbf{y}(t) = \begin{bmatrix} \theta(t) \\ \alpha(t) \end{bmatrix}; \begin{array}{l} \text{angle position of the arm} \\ \text{angle position of the pendulum} \end{array}$$

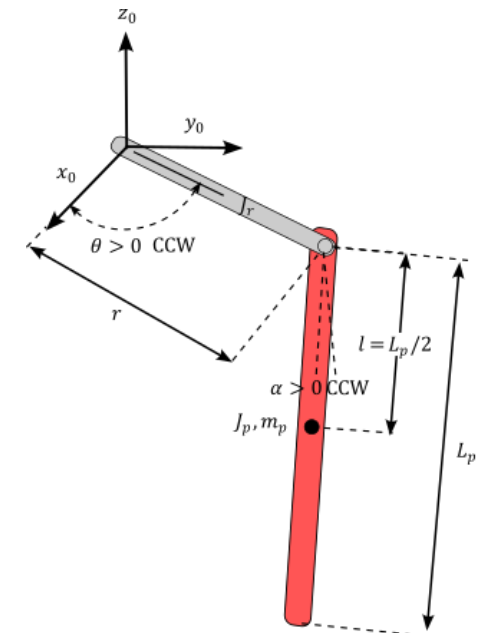


Fig. I.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

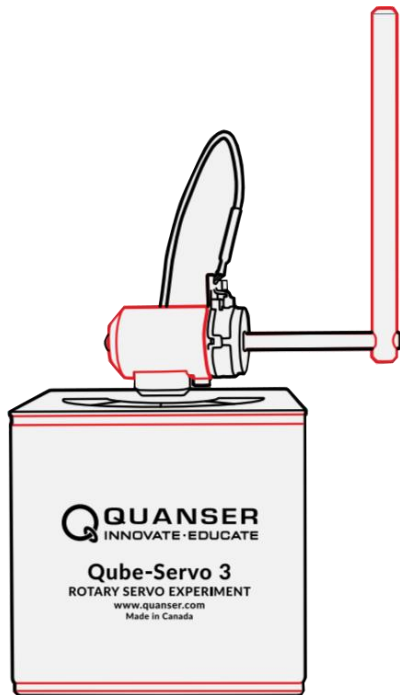
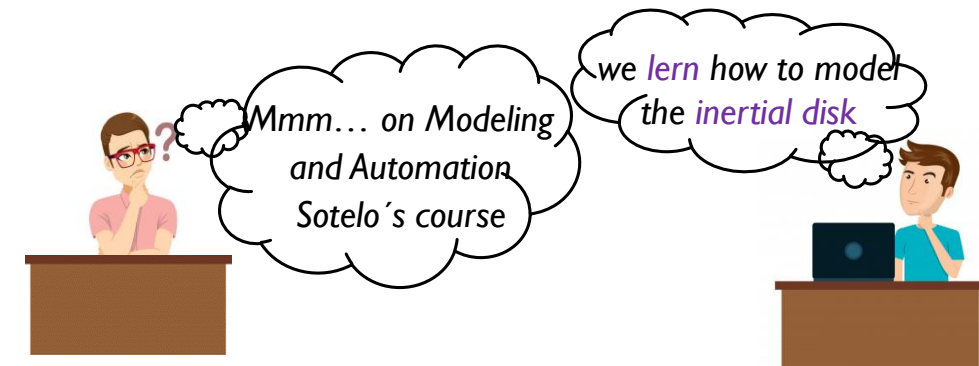


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# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

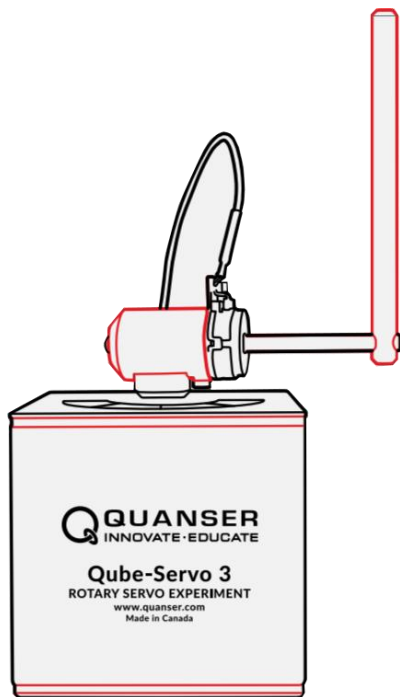
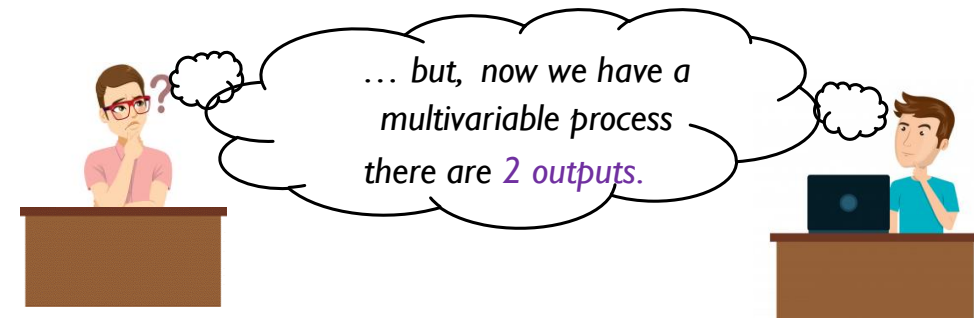
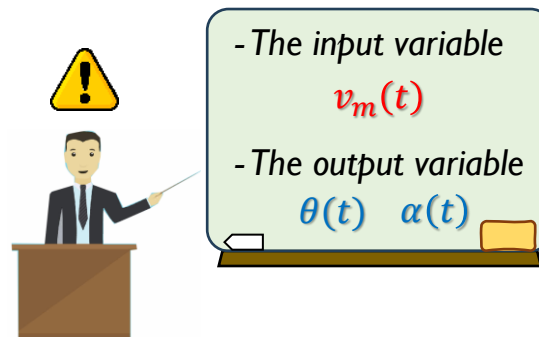


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### Example: Inverted Pendulum

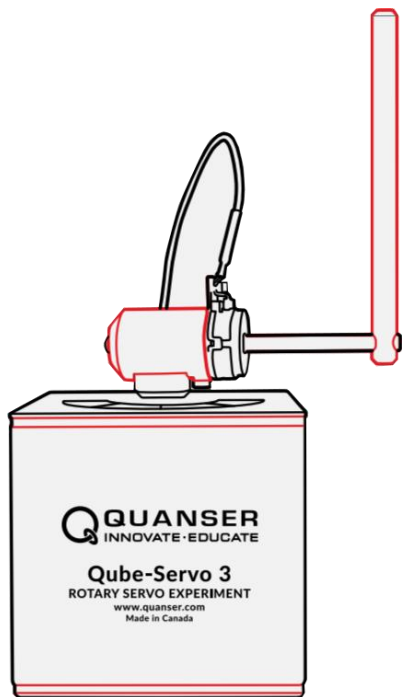
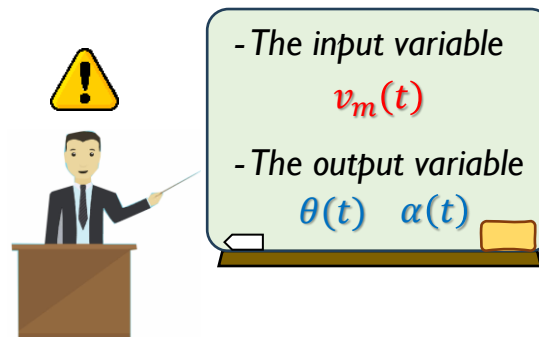


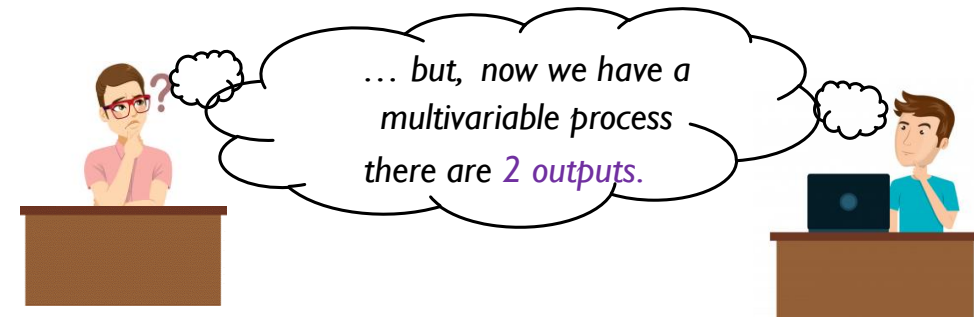
Fig. 1.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.
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You must consider that based on model, we have:



$$\begin{aligned} J_r \ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) &= \tau - b_r \dot{\theta}(t) \\ J_p \ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) &= -b_p \dot{\alpha}(t) \\ \tau &= \frac{k_m}{R_m} [v_m(t) - k_m \dot{\theta}(t)] \end{aligned}$$





# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

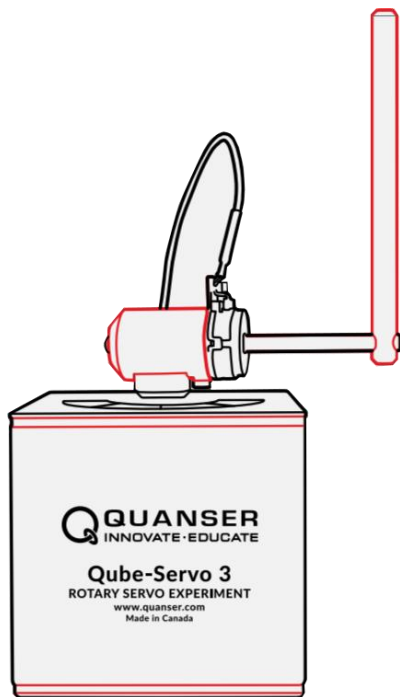


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- 1 Analyze the differential equation that describes the process behavior of the system.
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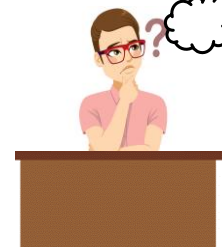
You must consider that based on model, we have:



- The input variable  
 $v_m(t)$   
- The output variable  
 $\theta(t)$   $\alpha(t)$

$$\begin{aligned} J_r \ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) &= \tau - b_r \dot{\theta}(t) \\ J_p \ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) &= -b_p \dot{\alpha}(t) \\ \tau &= \frac{k_m}{R_m} [v_m(t) - k_m \dot{\theta}(t)] \end{aligned}$$

$$\begin{aligned} \dot{x}(t) &= \mathbf{A}x(t) + \mathbf{B}u(t) \\ y(t) &= \mathbf{C}x(t) + \mathbf{D}u(t) \end{aligned}$$



# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

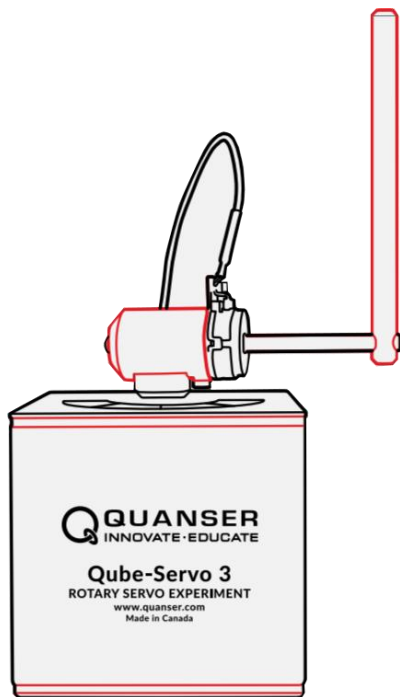


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.
- 2 Obtain the linear time invariant state space representation.

You must consider that based on model, we have:

**Step 1:** Define the state variables

$$\begin{aligned}x_1(t) &= \theta(t) \\x_2(t) &= \alpha(t) \\x_3(t) &= \dot{\theta}(t) \\x_4(t) &= \dot{\alpha}(t)\end{aligned}$$

**Step 2:** Determine  $\dot{x}(t)$

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \\ \dot{x}_4(t) \end{bmatrix} \begin{matrix} ? \\ ? \\ ? \\ ? \end{matrix}$$

$$\begin{aligned}J_r \ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) &= \tau - b_r \dot{\theta}(t) \\J_p \ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) &= -b_p \dot{\alpha}(t) \\\tau &= \frac{k_m}{R_m} [v_m(t) - k_m \dot{\theta}(t)]\end{aligned}$$

$$\begin{aligned}\dot{x}(t) &= \mathbf{A}x(t) + \mathbf{B}u(t) \\y(t) &= \mathbf{C}x(t) + \mathbf{D}u(t)\end{aligned}$$



# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

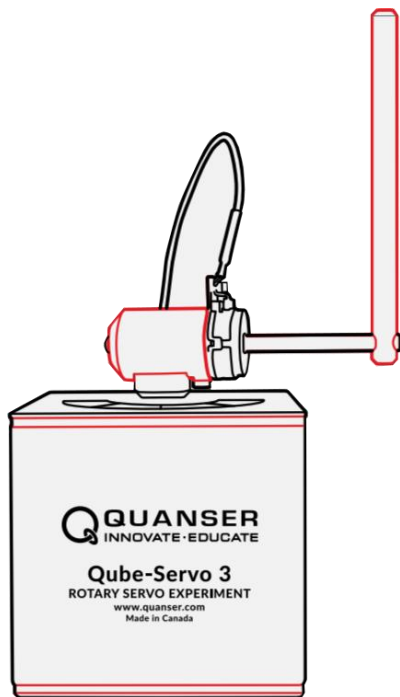


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.
- 2 Obtain the linear time invariant state space representation.
- 3 Validate your model using Quanser Interactive Labs.





## Practice 9 - State Feedback Gain Matrix

# **Practice 9** - State Feedback Gain Matrix

## **Part A** - State Space Modelling

## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling

## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling

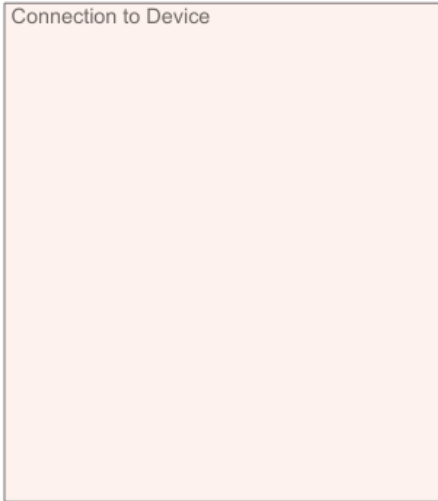




## Practice 9 - State Feedback Gain Matrix

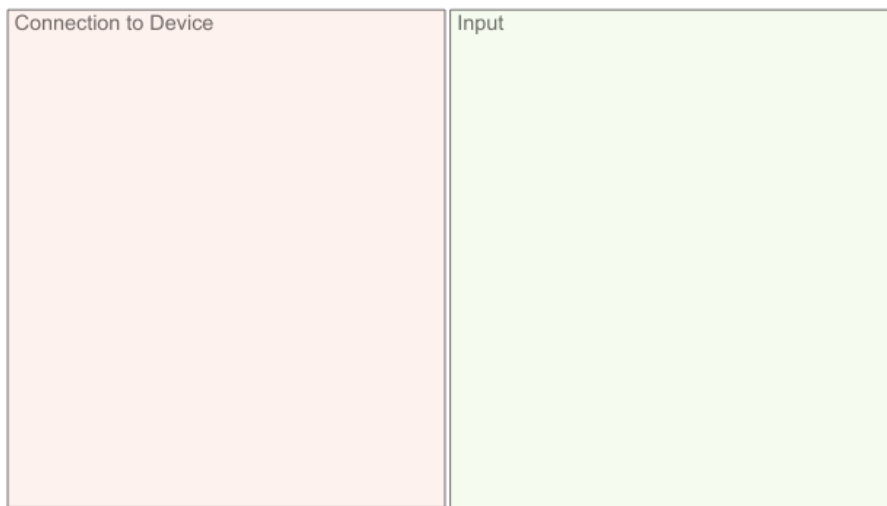
### Part A - State Space Modelling

Connection to Device



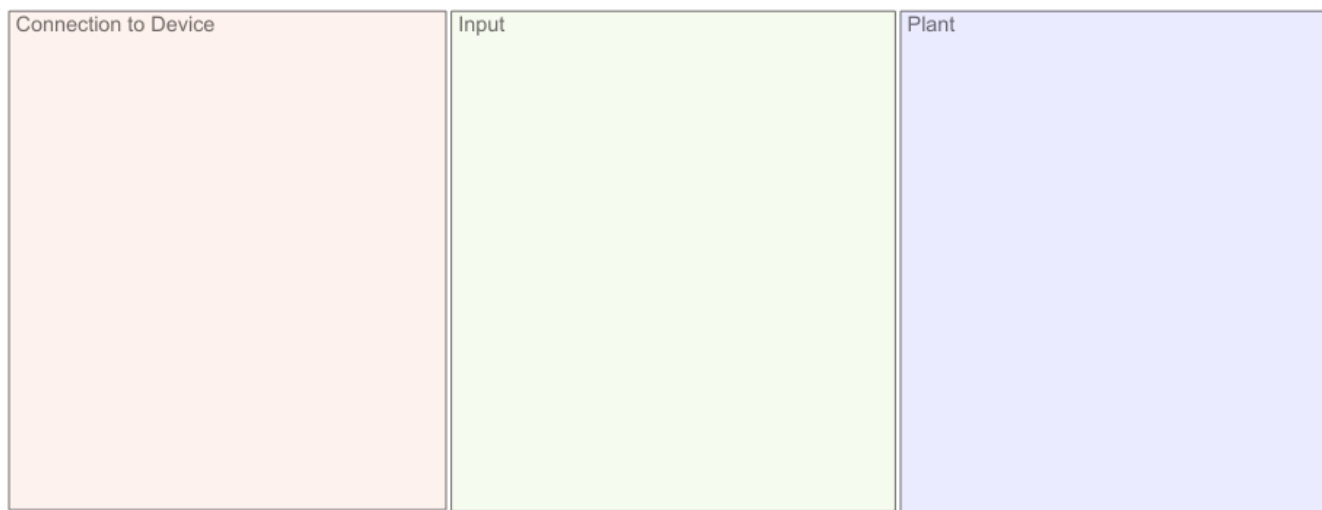
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

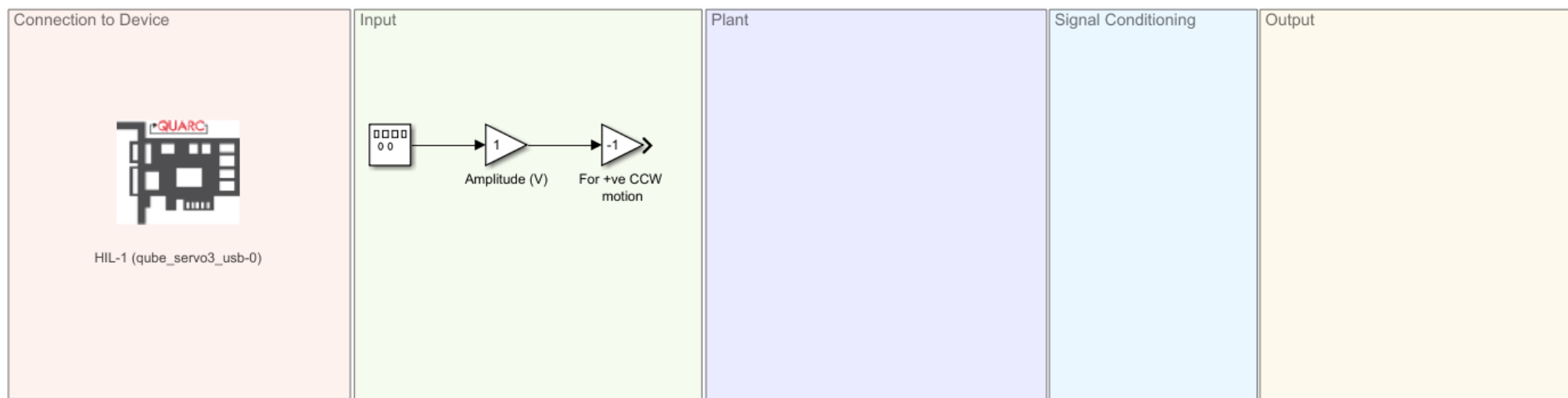
### Part A - State Space Modelling





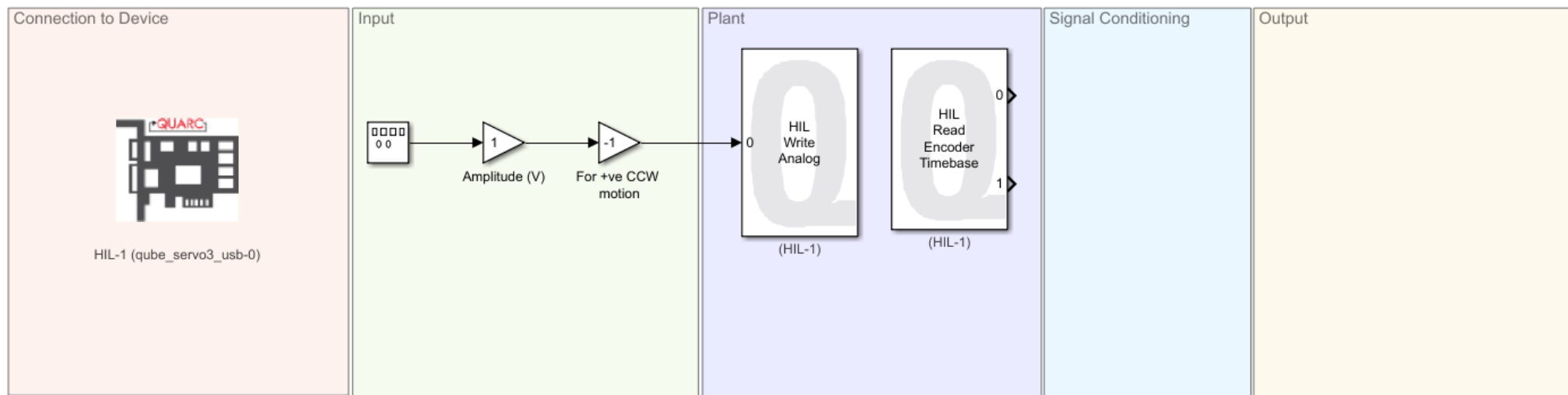
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



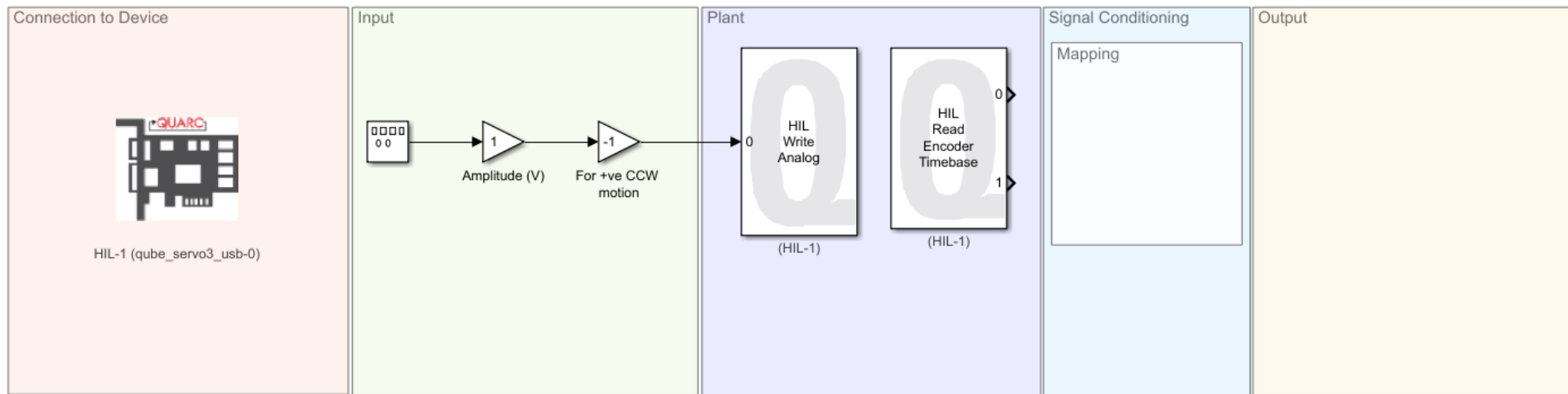
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



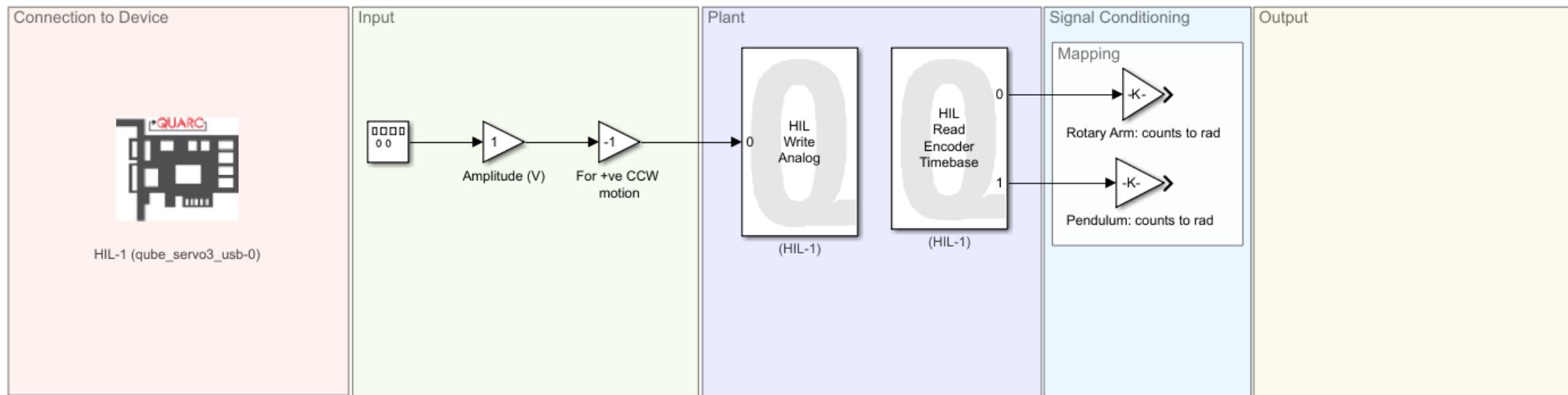
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



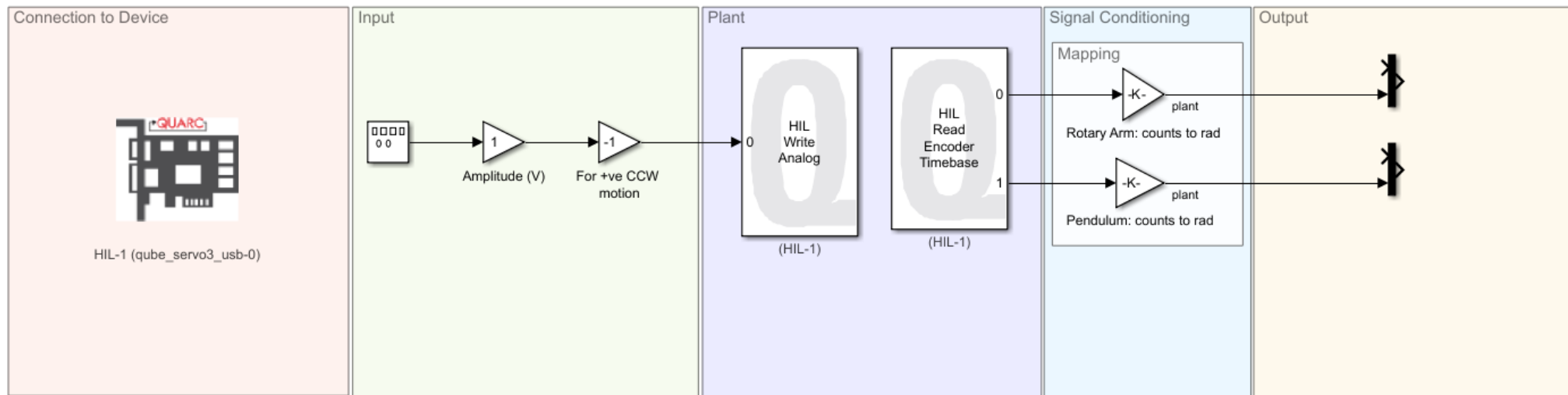
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



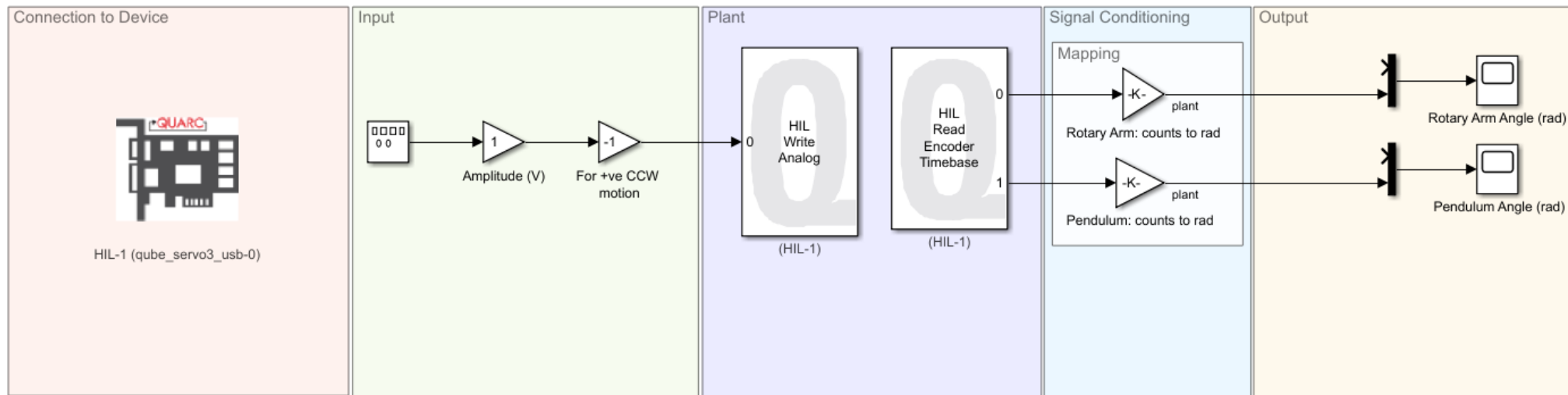
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



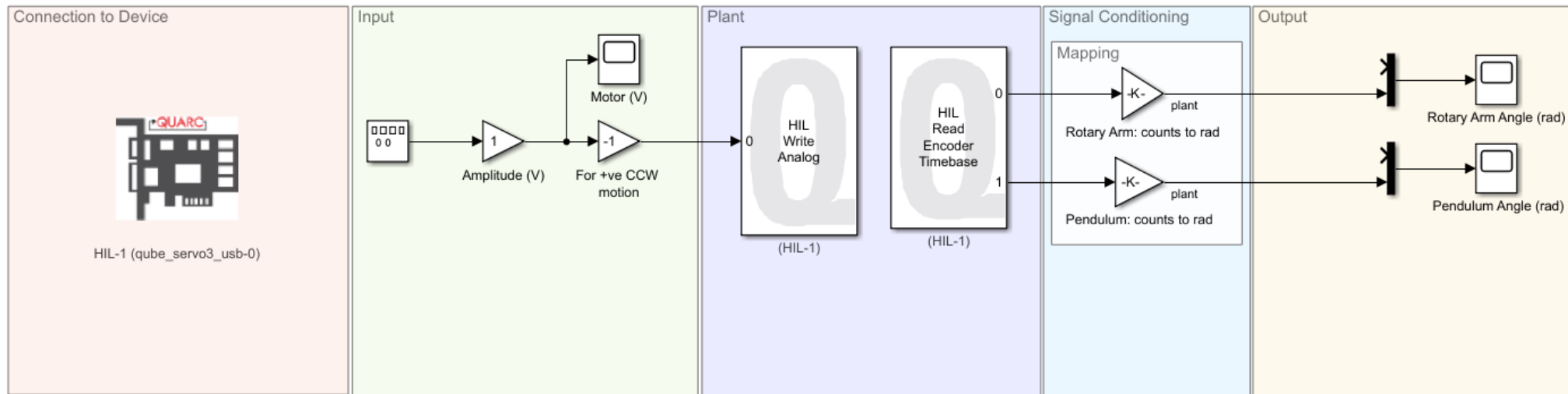
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



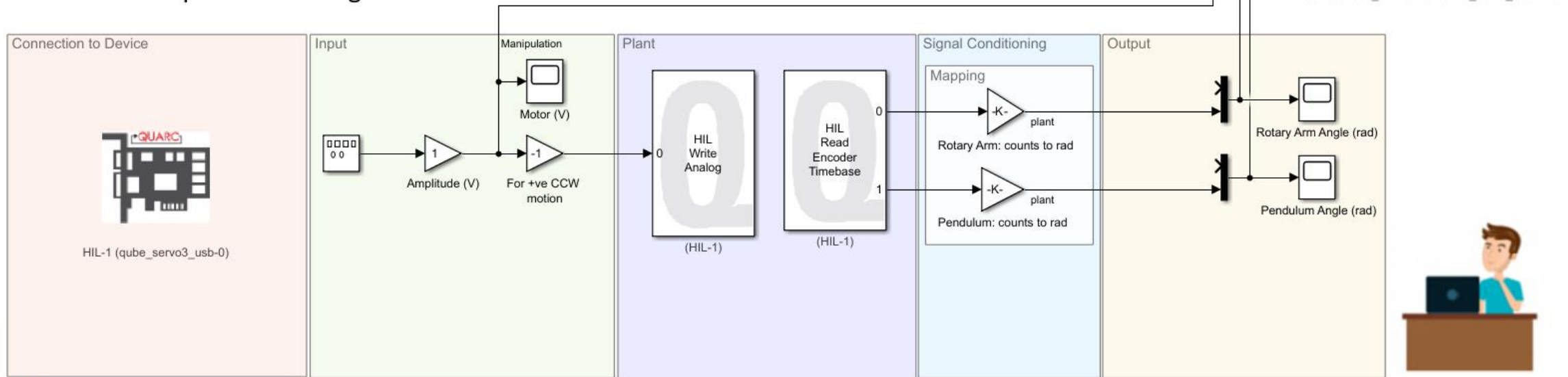
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



# Practice 9 - State Feedback Gain Matrix

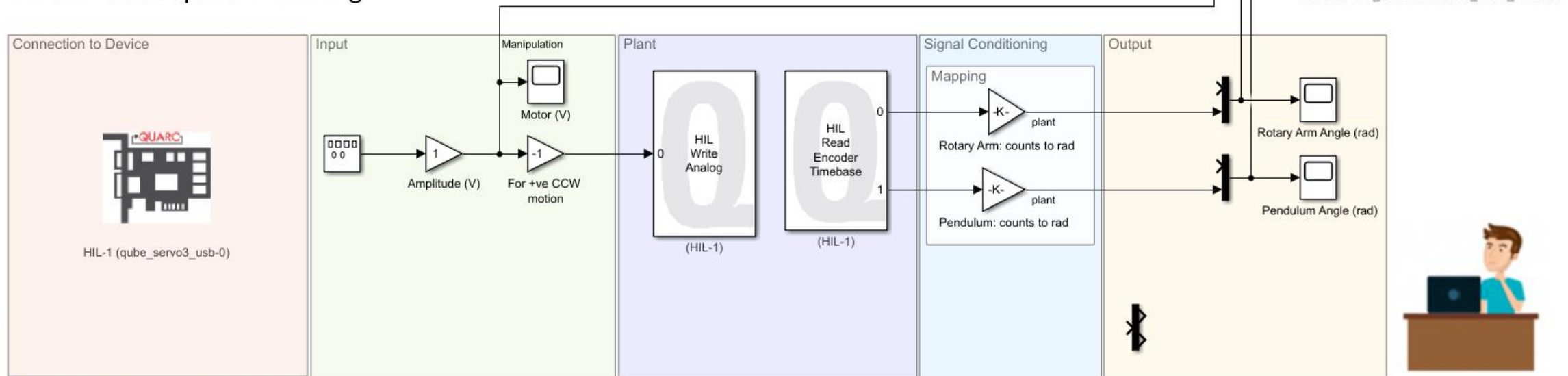
## Part A - State Space Modelling





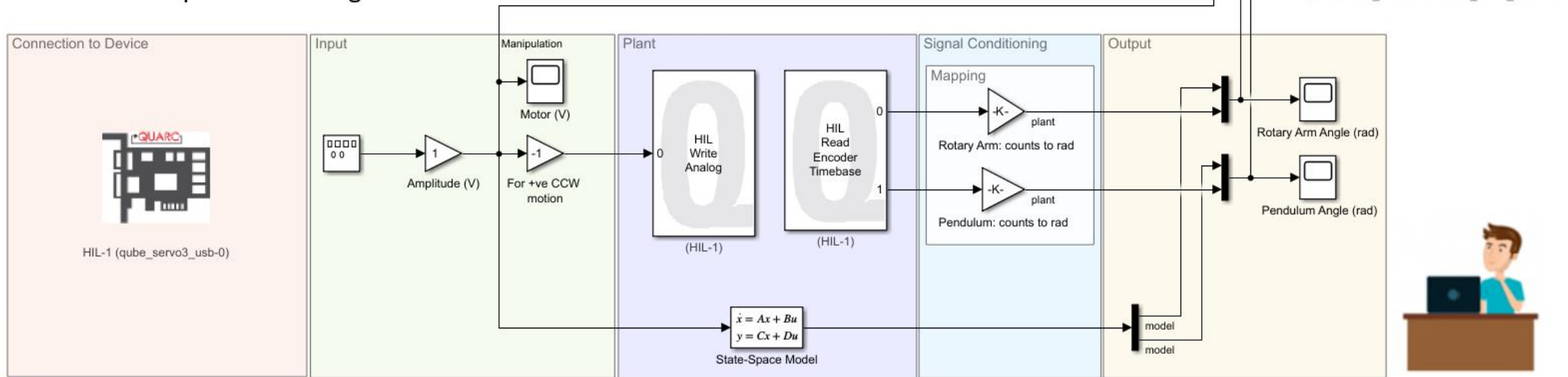
## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



## Practice 9 - State Feedback Gain Matrix

### Part A - State Space Modelling



# Practice 09

## Part A. State Space Modeling

### Example: Inverted Pendulum

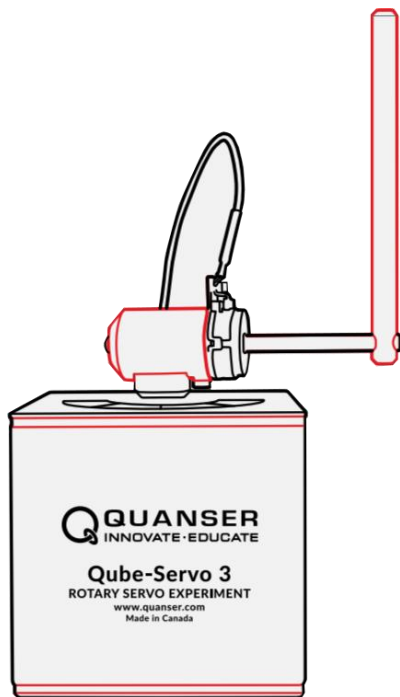


Fig. 1.1 Qube-Servo 3 Inverted pendulum.

- 1 Analyze the differential equation that describes the process behavior of the system.
- 2 Obtain the linear time invariant state space representation.
- 3 Validate your model using Quanser Interactive Labs.



# Thank you!

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